

VARUN SAXENA

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EDUCATION

University of Virginia | School of Arts and Sciences
M.S in Computer Science

GPA: 3.813
May 2027

SKILLS

Coding Proficiencies and Frameworks: Java, Python, Javascript, HTML, CSS, C, SQL, React.JS, PyTorch, TensorFlow, Flask
Knowledge/Technologies: Git, Github, Terminal, Visual Studio Code, Eclipse, MySQL, Firebase, Azure OpenAI, Microsoft 365

EXPERIENCE

ICF International | AI Software Developer Intern Jun. 2025-Aug. 2025

- Developed automated **data ingestion pipelines** to process government contract data for AI-driven proposal scoring systems
- Integrated **Azure AI Foundry** and **Python** workflows into **Microsoft 365**, enabling natural language parsing, document summarization, and classification
- Reduced manual processing time by **40%** through automation, increasing throughput and improving response accuracy across multiple departments
- Collaborated with business analysts to define KPIs for model performance and workflow efficiency

UVA School of Engineering | Computer Systems and Organizations 1 TA Jan. 2025-Present

- Assisted with design and refinement of course content, including lectures, homework, and exams
- Held office hours and assisted in in-person lectures to 100+ students, teaching the fundamentals of binary, circuits, machine code, assembly, and C programming
- Achieved class averages of over 80% on exams and over 85% on homework, demonstrating improved student understanding

Engage | Machine Learning and AI Intern Aug. 2024-Jan. 2025

- Preprocessed and categorized 10,000+ lecture transcripts and student comments using **NLTK**, **TF-IDF**, and **topic modeling**
- Built and evaluated classification models to cluster discussions by topic relevance, achieving **90% accuracy** on test data
- Deployed the pipeline across 15+ university courses (2,000+ students), leading to a **30% increase in engagement** and improved analytics for instructors

UVA Machine Learning and Artificial Intelligence | Research Assistant Aug. 2024-Jun. 2025

- Collected and cleaned open-source datasets to evaluate **generative AI model performance** across multiple benchmarks
- Designed **evaluation metrics** (accuracy, latency, cost per query) to balance efficiency and performance in LLM deployment
- Reduced model operation cost by **70%** while maintaining **benchmark accuracy**, demonstrating data-driven optimization and statistical analysis
- Analyzed dataset bias patterns using **Python (Pandas, NumPy, Matplotlib)** to identify correlation trends between model size and performance

PROJECTS

Infracool | Programming Lead Mar. 2025-Apr. 2025

- Led development of an **Urban Heat Island Risk Predictor**, combining **climate datasets** and **geospatial data (ZIP code-level)** to recommend mitigation strategies
- Engineered data pipelines with **Python (Flask, Pandas)** for real-time heat risk visualization and analytics dashboarding
- Awarded **1st place** among 10+ teams for data-driven social impact and technical excellence in the UVA Data Entrepreneurship Challenge

Portfolio Website | Designer and Creator Jun. 2024

- Engineered a personal portfolio website, accomplishing seamless and dynamic design elements
- Employed the React.JS framework, along with Javascript, CSS, and HTML to display an engaging frontend design

Nth Degree Robotics Team | Autonomous Lead Aug. 2022-Jun. 2023

- Conceptualized and created an image processing system, distinguishing different picture designs and recognizing color patterns, allowing for dynamic autonomous movement based off random pattern signals
- Applied OpenCV/TensorFlow Image Processing, allowing for a 50% accuracy increase in color recognition
- Allowed 50% increase in overall scoring in FTC robotic competitions, placing top 4 in Virginia for autonomous scoring

RELEVANT COURSEWORK

Classes: Data Structures and Algorithms 1 & 2, Computer Systems and Organizations 1 & 2, Discrete Mathematics 1 & 2, Introduction to Machine Learning, Software Development Essentials, Intro to Cybersecurity